

BEEE - UNIT -5

Syllabus: Rectifiers and power supplies: Block diagram description of a DC power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple zener voltage regulator.

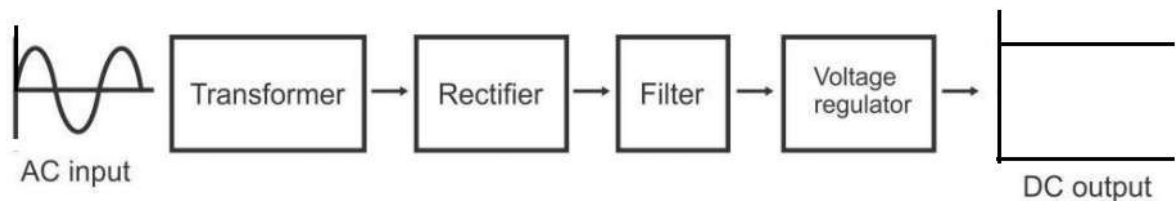
Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response.

Electronic Instrumentation: Block diagram of an electronic instrumentation system.

Short Questions:

- 1. Draw the block diagram of a DC power supply. (or) What are the blocks of a DC power supply?**

Ans:



- 2. What is the function of a capacitor filter in a rectifier circuit?**

Ans: A capacitor filter is used to reduce ripple in the output of a rectifier. It smooths the pulsating DC voltage by charging during the peak of the rectified voltage and discharging during the intervals, thereby providing a steadier DC output.

- 3. What is a Zener voltage regulator?**

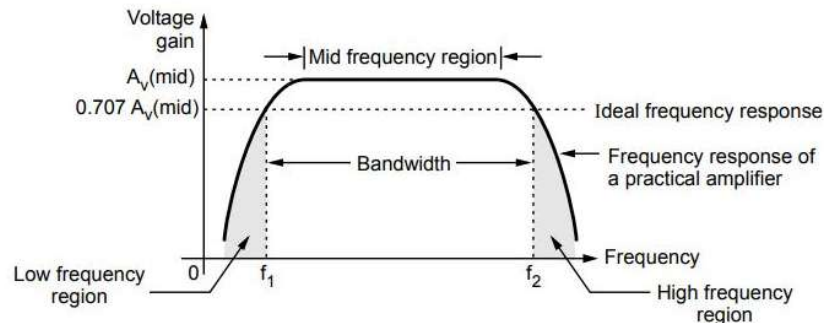
Ans: A Zener voltage regulator is a device that operates the Zener diode in reverse breakdown region to provide a stable and fixed DC output voltage across the load.

- 4. What is a Public Address (PA) system?**

Ans: A PA system is used to amplify and broadcast audio signals to a large audience.

5. What is the frequency response of a common emitter (RC coupled) amplifier?

Ans:



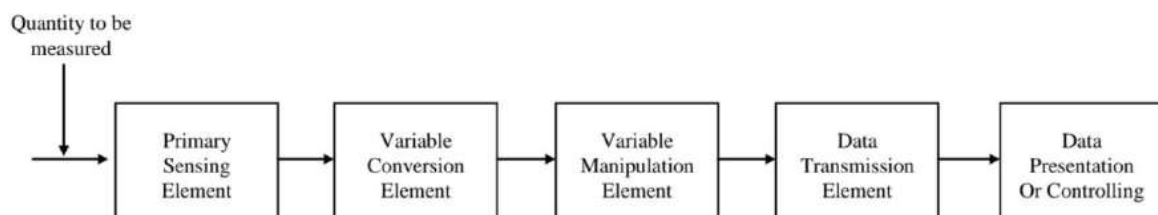
- The frequency response of an RC coupled amplifier shows how gain varies with frequency.
- The frequencies at which gain falls to 70.7% of maximum are called cut-off frequencies (f_L and f_H).
- The difference between them is called bandwidth.

6. What is an amplifier?

Ans: An amplifier is an electronic device that **increases the amplitude of a signal** (voltage, current, or power) without changing its original shape.

**7. Draw the block diagram of an electronic instrumentation system. (or)
What are the basic blocks of instrumentation system?**

Ans:



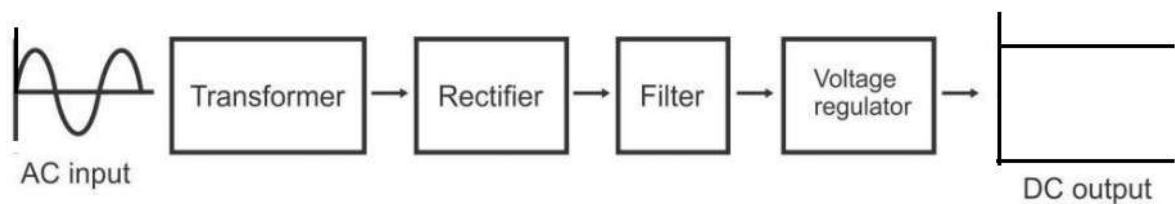
Long Questions:

1. Draw and explain the block diagram of a DC power supply in a neat and organized manner.

Ans: Electrical power is mostly generated, transmitted, and distributed in the form of AC (Alternating Current) because it is economical and easy to handle over long distances.

A DC power supply is essential because most electronic devices and circuits require a stable and constant DC voltage for proper operation.

A typical DC power supply consists of the following stages shown in Fig(a):



Fig(a): Block Diagram of DC Power Supply

Working of Each Block:

A typical DC power supply consists of a transformer, rectifier, filter, and voltage regulator connected in sequence.

Transformer: The AC input is first applied to the transformer, which steps the voltage up or down depending on the requirement of the electronic circuit. In addition to voltage transformation, the transformer also provides electrical isolation from the supply, which is an important safety feature and helps in reducing unwanted noise.

Rectifier: The output from the transformer is then fed to the rectifier. The rectifier converts the AC voltage into a pulsating DC voltage by eliminating the negative portion of the waveform or by converting both halves of the AC signal into unidirectional current. However, this pulsating DC is not suitable for most electronic applications because it contains ripples.

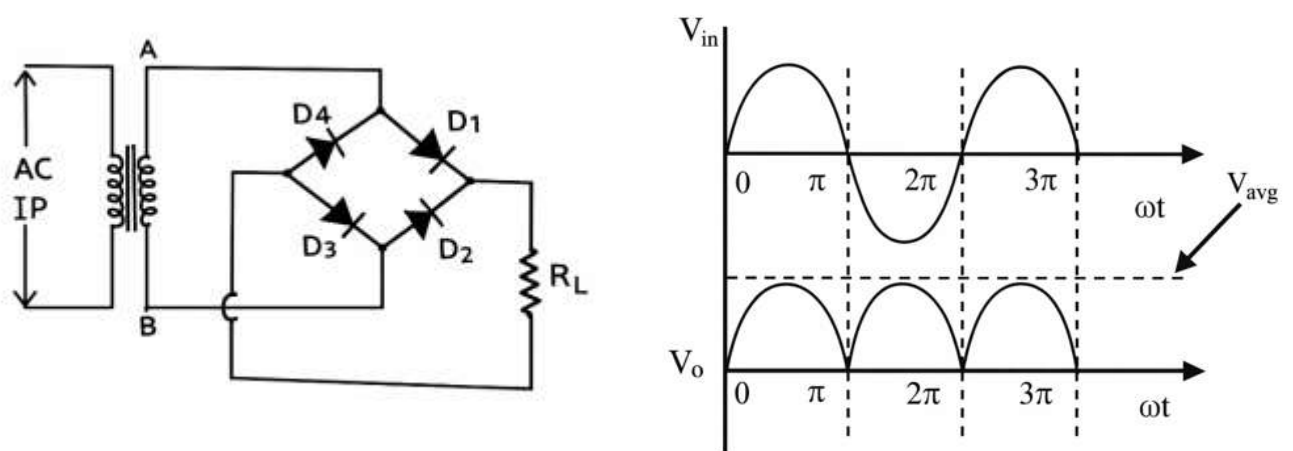
Filter: To reduce ripples in the output of the rectifier is passed through a filter circuit. The filter smooths the pulsating DC and produces a more uniform DC voltage.

Voltage Regulator: The voltage regulator maintains a constant output voltage irrespective of variations in the input supply or changes in the load conditions. Thus, it ensures a stable and steady DC output.

2. Explain the working of a full wave bridge rectifier.

Ans: A full wave bridge rectifier is a circuit used to convert alternating current (AC) into direct current (DC) by utilizing both the positive and negative half cycles of the input waveform.

It consists of four diodes connected in a bridge configuration, along with a load resistor across which the output is taken. This arrangement allows current to flow through the load in the same direction during both halves of the AC cycle.



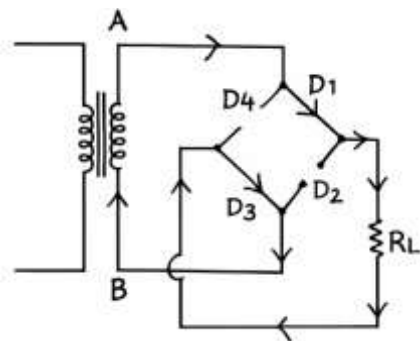
Fig(b): Full wave Rectifier with wave forms

When an AC voltage is applied to the input terminals of the bridge, the operation of the circuit can be understood by considering the two half cycles separately.

During the positive half cycle of the input, point A is more positive than point B. Thus, diodes D1 and D3 are forward biased, while diodes D2 and D4 are reverse biased.

Now, the current flows through:

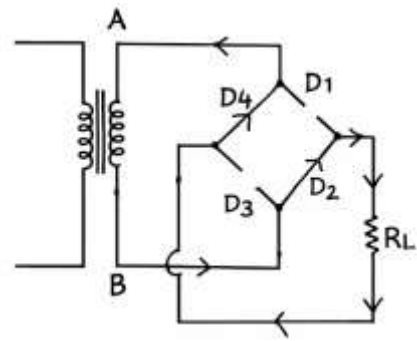
$A \rightarrow D1 \rightarrow R_L \rightarrow D3 \rightarrow B \rightarrow A$



During the negative half cycle of the input, point B is more positive than point A. Thus, diodes D2 and D4 are forward biased, and diodes D1 and D3 are reverse biased.

Now, the current flows through:

B → D2 → RL → D4 → A → B



In both the cases, the current flows through R_L in the same direction. As a result, the output obtained across the load is a pulsating DC voltage, which contains ripples. These ripples can be reduced by using a filter circuit to obtain a smoother DC output. The main advantage of a full wave bridge rectifier is that it utilizes both half cycles of the input signal, resulting in higher efficiency and better output compared to a half wave rectifier.

Average and RMS values of output of the Full Wave rectifier is

$$V_{avg} = \frac{2V_m}{\pi} \qquad V_{rms} = \frac{V_m}{\sqrt{2}}$$

Full wave bridge rectifier with Filter: When AC voltage is applied to the input, the bridge rectifier converts both the positive and negative half cycles of the AC waveform into pulsating DC. During each half cycle, two diodes conduct while the other two remain reverse biased, ensuring that current flows through the load in the same direction. Thus, the output obtained is a full wave rectified voltage.

However, this output still contains ripples. To reduce these ripples, a filter capacitor is connected across the load. During the peak of the rectified voltage, the capacitor charges up to the maximum value. When the input voltage decreases, the capacitor starts discharging slowly through the load, thereby maintaining the output voltage. This process smooths the pulsating DC and reduces ripple content.

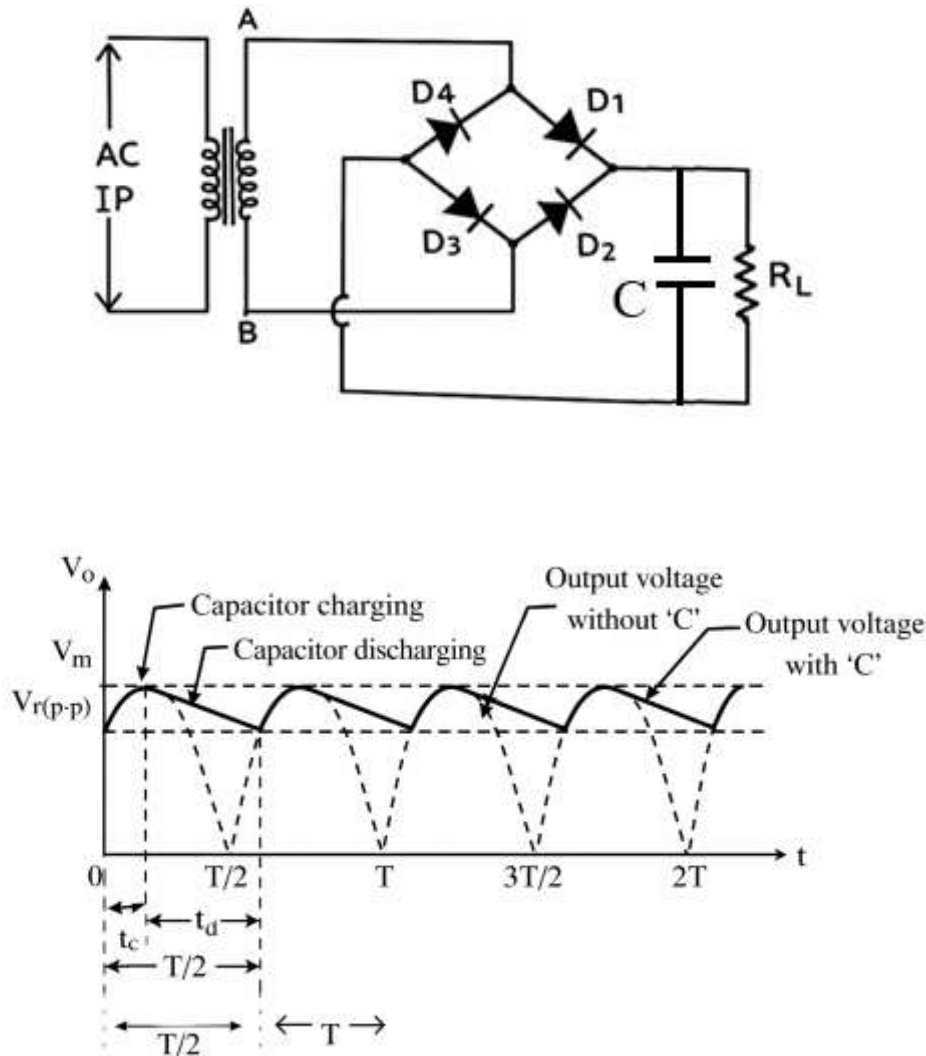


Fig.(c): Full wave rectifier with capacitor and corresponding wave forms

3. What is the function of a capacitor filter in a rectifier circuit?

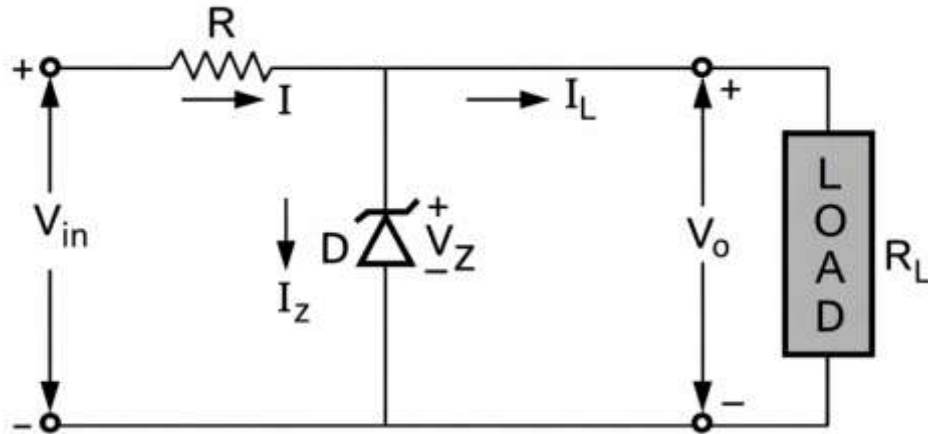
Ans: A capacitor filter in a rectifier circuit is used to reduce the ripple content present in the rectified output and to provide a smooth DC voltage.

When the rectifier produces pulsating DC, the capacitor charges up to the peak value of the output voltage during the conduction period. As the input voltage starts to decrease, the capacitor discharges slowly through the load, maintaining the output voltage nearly constant. This charging and discharging action fills the gaps between the pulses and reduces voltage fluctuations.

Thus, the main function of a capacitor filter is to smooth the pulsating DC output and improve the quality of the DC supply.

4. Explain the working of simple Zener voltage regulator.

Ans: A simple Zener voltage regulator is used to maintain a constant output voltage despite variations in input voltage or load conditions. It consists of a Zener diode connected in reverse bias across the load, along with a series resistor connected to the supply.



Fig(d): Zener voltage regulator

When the input voltage is applied, the Zener diode operates in its breakdown region if the voltage across it reaches the Zener voltage. In this region, the Zener diode maintains a constant voltage across itself, which is equal to its breakdown voltage. Since the load is connected in parallel with the Zener diode, the output voltage across the load also remains constant.

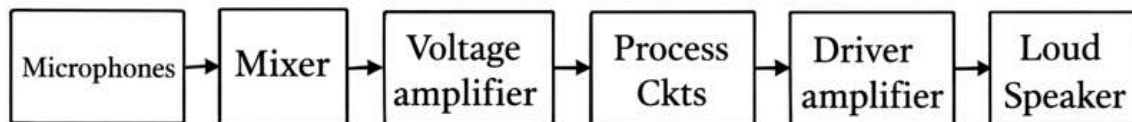
If the input voltage increases, the current through the series resistor increases. The Zener diode conducts more current, thereby maintaining the same voltage across the load. Conversely, if the input voltage decreases, the Zener current reduces, but the voltage across the load remains nearly constant as long as the diode stays in the breakdown region.

Similarly, when the load current changes, the Zener diode automatically adjusts its current to keep the output voltage constant. Thus, the Zener diode acts as a voltage stabilizer.

In this way, a simple Zener voltage regulator provides a **constant and regulated DC output voltage**, making it suitable for protecting electronic circuits from voltage variations.

5. Describe the block diagram of Public Address system.

Ans: A Public Address (PA) system is designed to amplify sound such as human voices or music and distribute it to a large audience over a wide area. It converts sound energy into electrical signals, processes and amplifies them, and converts them back into acoustic sound.



Fig(e): Block Diagram of Public Address System

A block diagram of a typical PA system includes the following components:

Microphone: A microphone is a transducer that converts sound waves into electrical signals of the same frequency. The strength of the signal depends on the sound level. In P.A. systems, usually more than one microphone is used along with an extra input for devices like tape or record players.

Mixer: The outputs from microphones and other audio sources are fed to the mixer. Its function is to combine different signals and keep each channel isolated before sending them to the main amplifier. Each channel has its own pre-amplifier and gain control, and then all signals are combined and amplified together.

Voltage Amplifier: It further amplifies the output of the mixer.

Processing Circuit: These circuits include master gain control and tone control circuits (Bass and Treble controls).

- **Bass Control:** Bass control permits cutting and boosting of low frequencies.
- **Treble Control:** Treble control permits cutting and boosting of high frequencies.
- **Master Control:** This acts as a gate for overall adjustment of the volume of all input (mixed) signals.

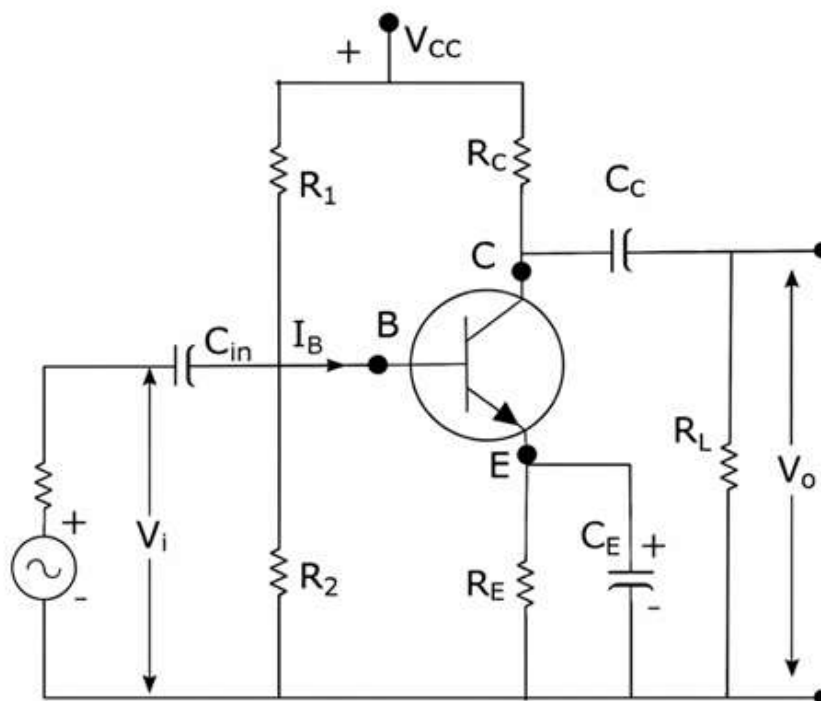
Driver Amplifier: This amplifier acts as a driver for the power amplifier and provides voltage amplification.

Power Amplifier: It increases the power of the mixed signal to the required level to drive the loudspeaker. It usually uses a push-pull circuit to reduce distortion and avoid transformer saturation. The output is given to the loudspeaker through a matching transformer for maximum power transfer.

Loudspeaker: Loudspeaker is a transducer which converts electrical signals of audio frequency into sound waves of the same frequency.

**6. Explain in detail about common emitter (RC coupled) amplifier. (or)
What is the frequency response of a common emitter (RC coupled) amplifier?**

Ans: A common emitter (CE) RC coupled amplifier is a widely used transistor amplifier in which the input is applied to the base, the output is taken from the collector, and the emitter is common to both input and output. The stages are connected using resistor-capacitor (RC) coupling, which is suitable for amplifying audio frequency signals.



Fig(f): Circuit diagram of RC Coupled amplifier

Circuit Description: The circuit consists of a transistor, biasing resistors, coupling capacitors, and load resistors.

The input signal is applied to the base through a coupling capacitor C_{in} , which blocks DC and allows only AC signals. The transistor is properly biased using a voltage divider so that it operates in the active region.

The collector is connected to the supply through a resistor R_C , and the emitter has a resistor R_E for stability. A bypass capacitor C_E is connected across the emitter resistor to increase the gain. The output is taken from the collector through another coupling capacitor C_C and supplied to the load.

Working: When an AC signal is applied to the input, it causes small variations in the base current. Due to transistor action, these small changes produce large variations in collector current.

During the positive half cycle of the input signal, the base current increases, which increases collector current and causes a larger voltage drop across the collector resistor. This results in a decrease in collector voltage.

During the negative half cycle, the base current decreases, reducing the collector current and increasing the collector voltage. Thus, the output voltage is amplified and is 180° out of phase with the input signal.

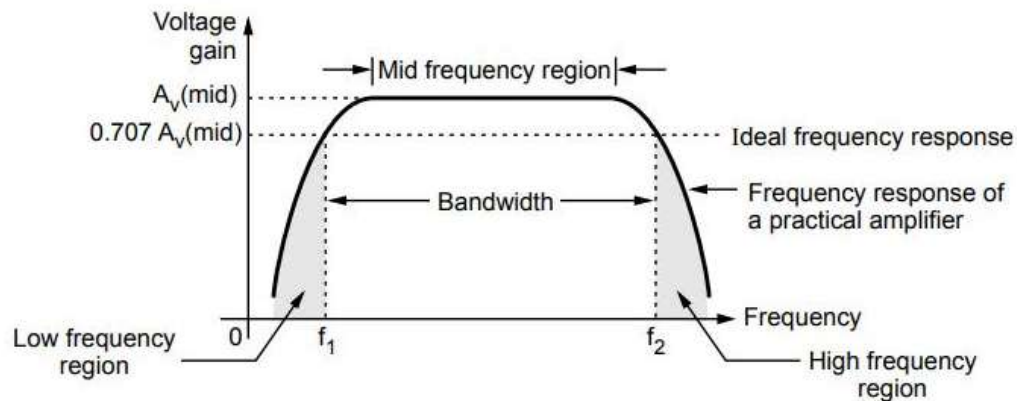
The RC coupled common emitter amplifier is an important amplifier circuit that provides significant amplification of weak AC signals with phase inversion, making it suitable for audio and communication applications.

Frequency Response:

The frequency response of an RC coupled amplifier shows how gain varies with frequency:

- a) Low Frequency Region: Gain is low due to the effect of coupling and bypass capacitors.
- b) Mid Frequency Region: Gain is maximum and nearly constant.
- c) High Frequency Region: Gain decreases due to internal transistor capacitances.

The frequencies at which gain falls to 70.7% of maximum are called cut-off frequencies (f_L and f_H). The difference between them is called bandwidth.

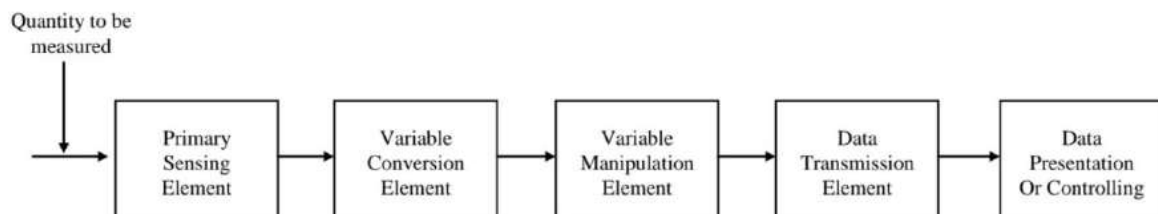


Advantages:

1. Provides good voltage gain
2. Simple and economical
3. Widely used for audio amplification

7. Explain the block diagram of an electronic instrumentation system.

Ans: An electronic instrumentation system is used to measure physical quantities such as temperature, pressure, and displacement, and convert them into a readable form.



Fig(g): Block Diagram of Electronic Instrumentation System

Block diagram shows a basic instrumentation system, it consists of primary sensing element, variable, manipulation element, data transmission, data presentation element.

(i) Primary sensing element: The primary sensing element is also known as sensor. Basically, transducers are used as a primary sensing element. Here, the physical quantity (such as temperature, pressure etc.,) are sensed and then converted into analogous signals.

(ii) Variable conversion element: It converts the output of primary sensing element into suitable form without changing the information. Basically these are secondary transducers.

(iii) Variable manipulation element: The output of transducer may be electrical signal i.e., voltage, current or other electrical parameter. Here, manipulation means change in numerical value of the signal. This element is used to convert the signal into suitable range.

(iv) Data Transmission Element: In some cases, it is not possible to measure and display the value at the same place (for example, temperature measurement in a furnace). So, the data is sent from one location to another through a transmission path called a data transmission element. Common transmission paths include pneumatic pipes, electrical cables, and radio links. When radio links are used, the system is called a telemetry system.

(v) Data presentation (or) Controlling element: Finally, the output is recorded or given to the controller to perform action. It performs different functions like indicating, recording or controlling.